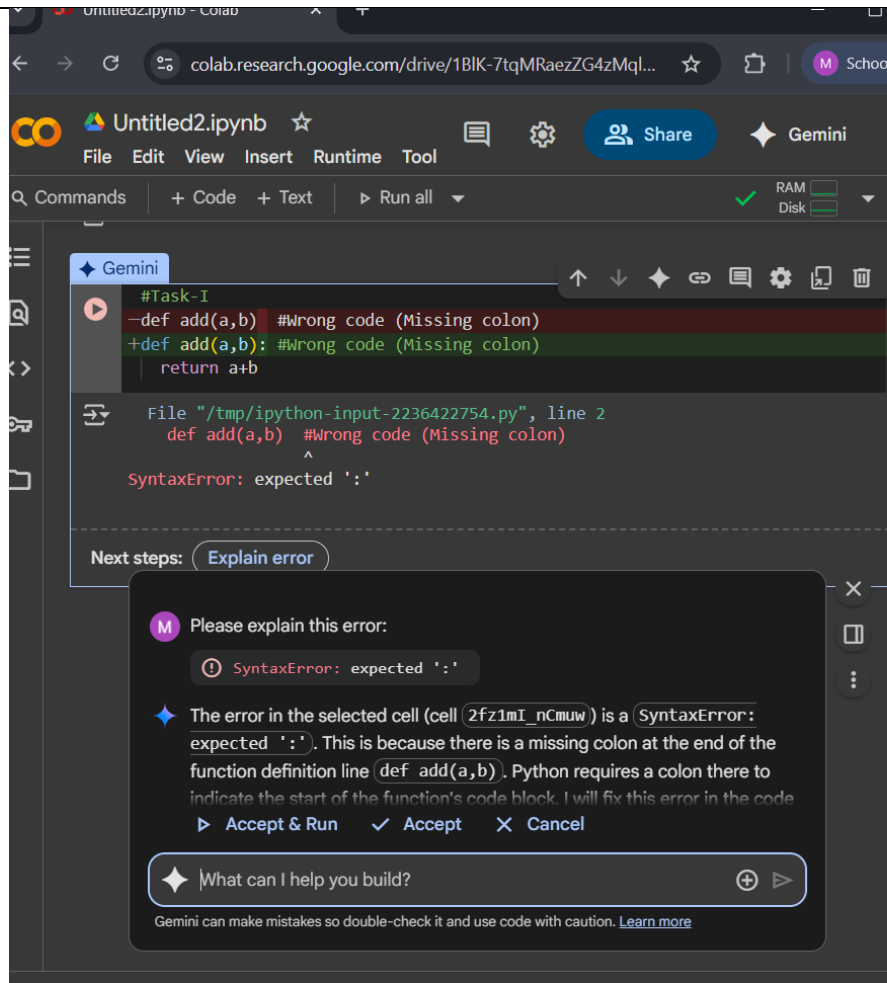


LAB ASSIGNMENT -7

Name: M.Preetham		
Roll no:2403A53030		
Batch No: CYB02		
Q.No.	Question	<i>Expected Time to complete</i>
1	Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs Lab Objectives: - To identify and correct syntax, logic, and runtime errors in Python programs using AI tools. - To understand common programming bugs and AI-assisted debugging suggestions. - To evaluate how AI explains, detects, and fixes different types of coding errors. - To build confidence in using AI to perform structured debugging practices. Lab Outcomes (Los): After completing this lab, students will be able to: - Use AI tools to detect and correct syntax, logic, and runtime errors. - Interpret AI-suggested bug fixes and explanations. - Apply systematic debugging strategies supported by AI-generated insights. - Refactor buggy code using responsible and reliable programming patterns. Task Description#1 - Paste a function with a missing colon (add(a, b)), and let AI fix the syntax error. <pre>python def add(a, b) return a + b</pre> Expected Output#1 - Corrected function with syntax fix	Week4 – Wednesday



Task Description#2 (Loops)

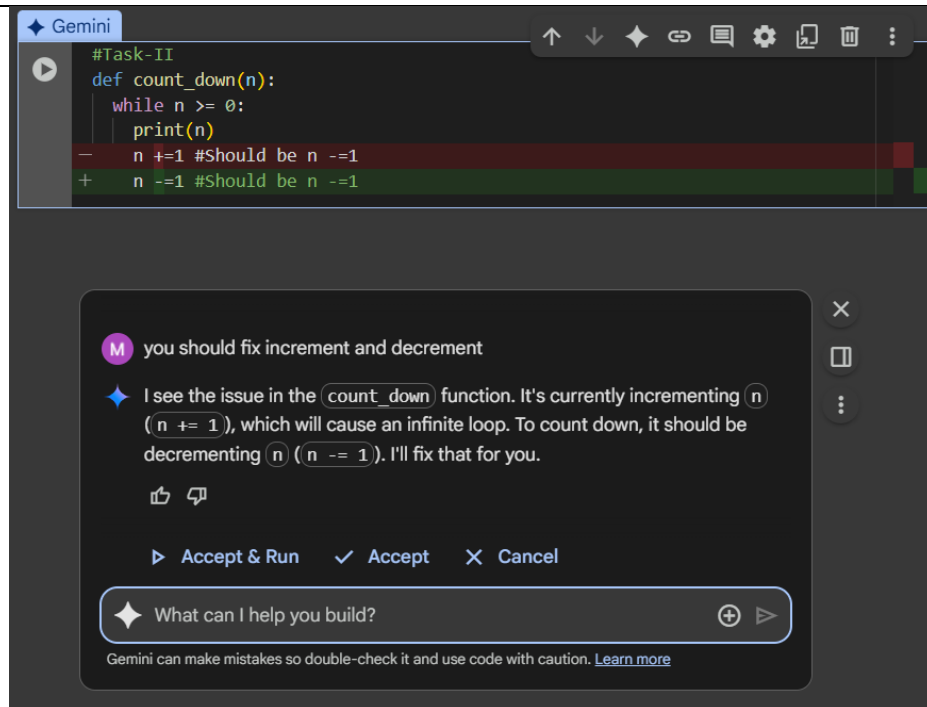
- Identify and fix a logic error in a loop that causes infinite iteration.

```
python

def count_down(n):
    while n >= 0:
        print(n)
        n += 1 # Should be n -= 1
```

Expected Output#2

- AI fixes increment/decrement error



Task Description#3

- Debug a runtime error caused by division by zero. Let AI insert try-except.

```
# Debug the following code
def divide(a, b):
    return a / b

print(divide(10, 0))
```

Expected Output#3

- Corrected function with safe error handling

Gemini

#Task-III

def divide(a, b):

- return a/b

print(divide(10,0))

+ try:

+ return a/b

+ except ZeroDivisionError:

+ print("Error: Cannot divide by zero!")

+ return None # Or you could return a specific value like float('inf') or a

+ print(divide(10,0))

M corrected funtion with safe error hadling

<> Empty cell

✦ You're right, the current code in cell (yYT6aZBB9J4d) will cause a `ZeroDivisionError` when trying to divide by zero. I will modify the cell to include error handling for this case using a `try-except` block.

▶ Accept & Run

✓ Accept

✕ Cancel

✦ What can I help you build?

⊕ ▶

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

Variables

Terminal

11:47 AM

Python 3

Task Description#4

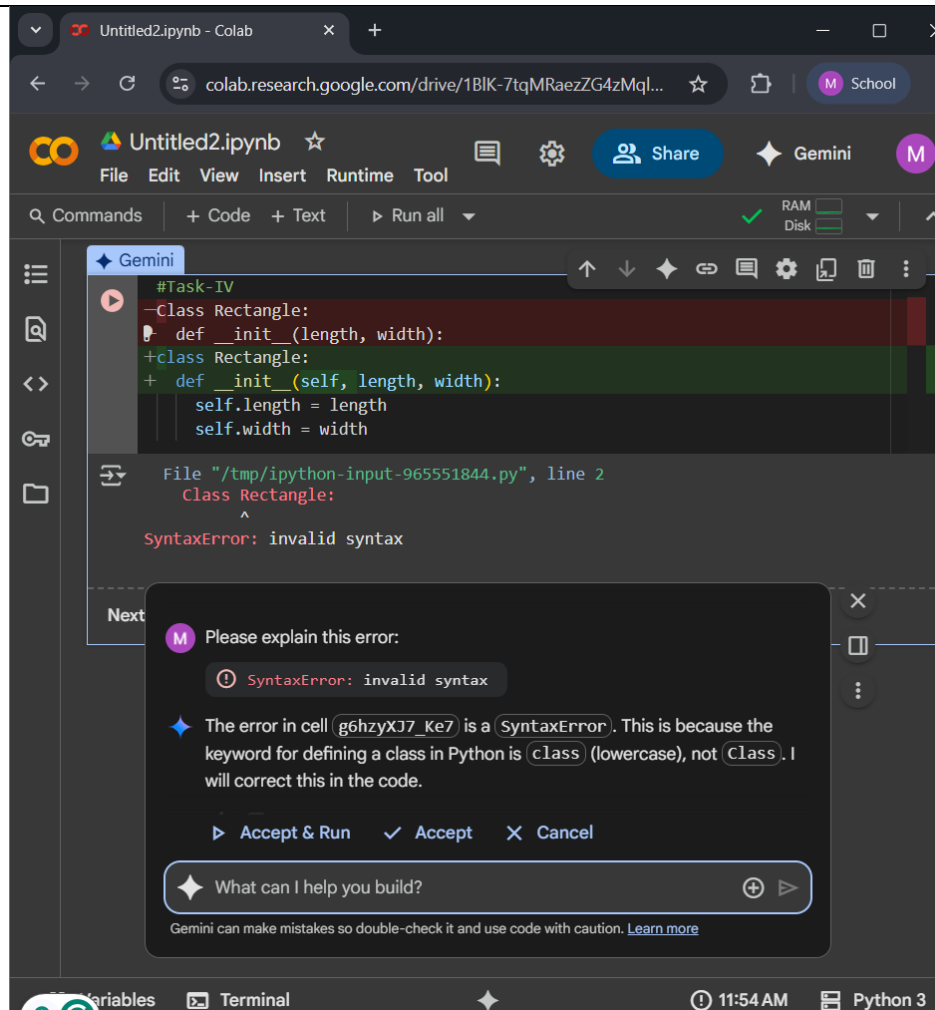
- Provide a faulty class definition (missing self in parameters). Let AI fix it

```
python

class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

Expected Output#4

- Correct `__init__()` method and explanation



Task Description#5

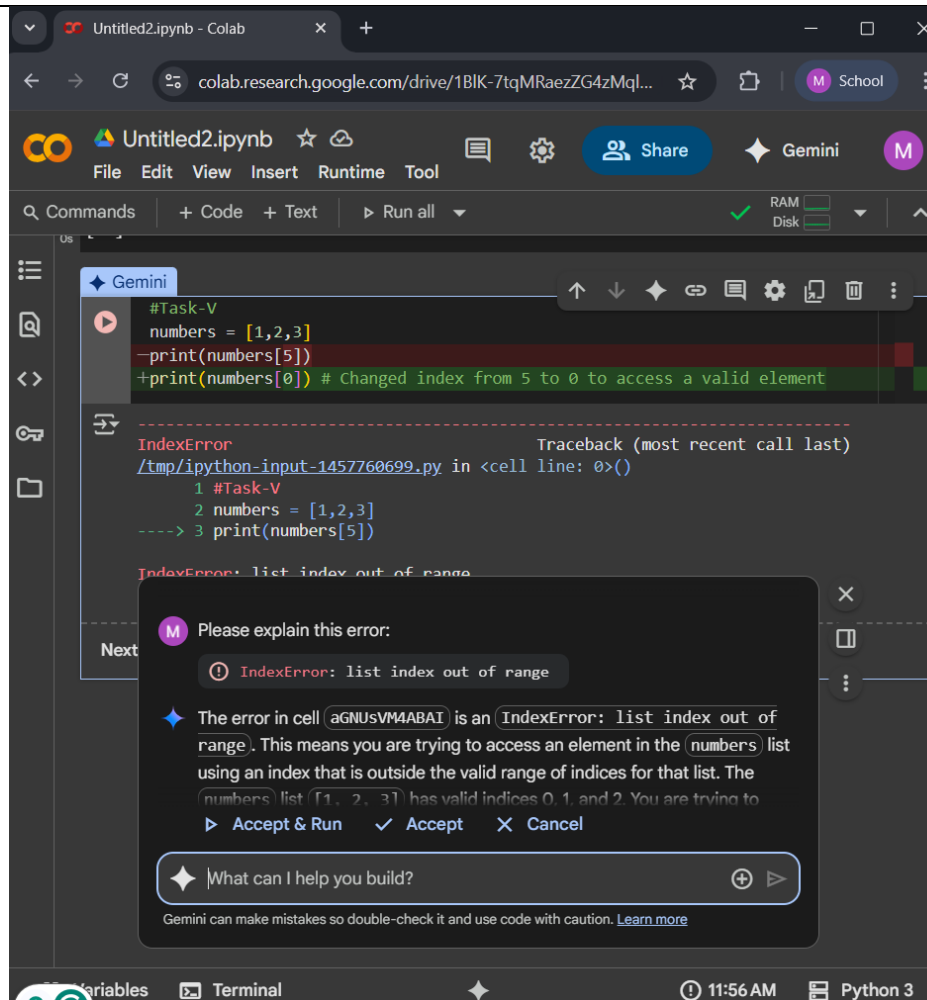
- Access an invalid list index and use AI to resolve the Index Error.

```
python

numbers = [1, 2, 3]
print(numbers[5])
```

Expected Output#5

- AI suggests checking length or using safe access logic



Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Identification of bugs	0.5
Application of AI-suggested fixes	0.5
Explanation and understanding of errors	0.5
Corrected code functionality	0.5
Report structure and reflection	0.5
Total	2.5 Marks